Week 5 — Confidence intervals

Business Forecasting

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My code (.R) Notes

Where we are

Describing data → Reasoning under uncertainty → Regression → Time series forecasting

Reasoning under uncertainty. A sample mean is a single number. This week we put a range of plausible values around it — a confidence interval.

- This week (W5): confidence intervals — for a mean (normal & Student's t) and for a variance (chi-square).

Fast & hands-on: every interval we derive, you build in R with `qnorm` / `qt` / `qchisq` — the exercises run right here in the deck, in your browser.

Confidence Intervals

Confidence intervals

- We calculated the mean price in our sample
- How confident are we that our estimate is close to the true parameter?
- A confidence interval measures the uncertainty around the estimate — a range of plausible values for the parameter.

Confidence intervals — a first look

- Our running sample this week: the 100 Mexico City Airbnb listings with a cleanliness score above 4.5. Their mean nightly price: 1245.43 pesos.
- Is it reasonable to think the true average price is 1100? What about 2000?
- Suppose we computed the confidence interval to be:

$$(1086.64,\ 1404.22)$$

- Where do these numbers come from?
    1. The sampling distribution of the sample mean tells us how far a point estimate typically lands from the true mean.
    2. The confidence interval uses that to say where the true mean plausibly lies.

The route this week

1. From the sample mean to the margin of error — how far a sample mean typically lands from the truth.
2. Building the interval — the recipe that turns $\bar{x}$, s and n into two endpoints.
3. What moves the width — confidence level, sample size, spread.
4. What “95% confident” means — what the 95% is a probability of.
5. Small samples: Student’s t — the fix when n is small and σ is unknown.

Then an appendix (an interval for the variance) and ten exam-style problems.

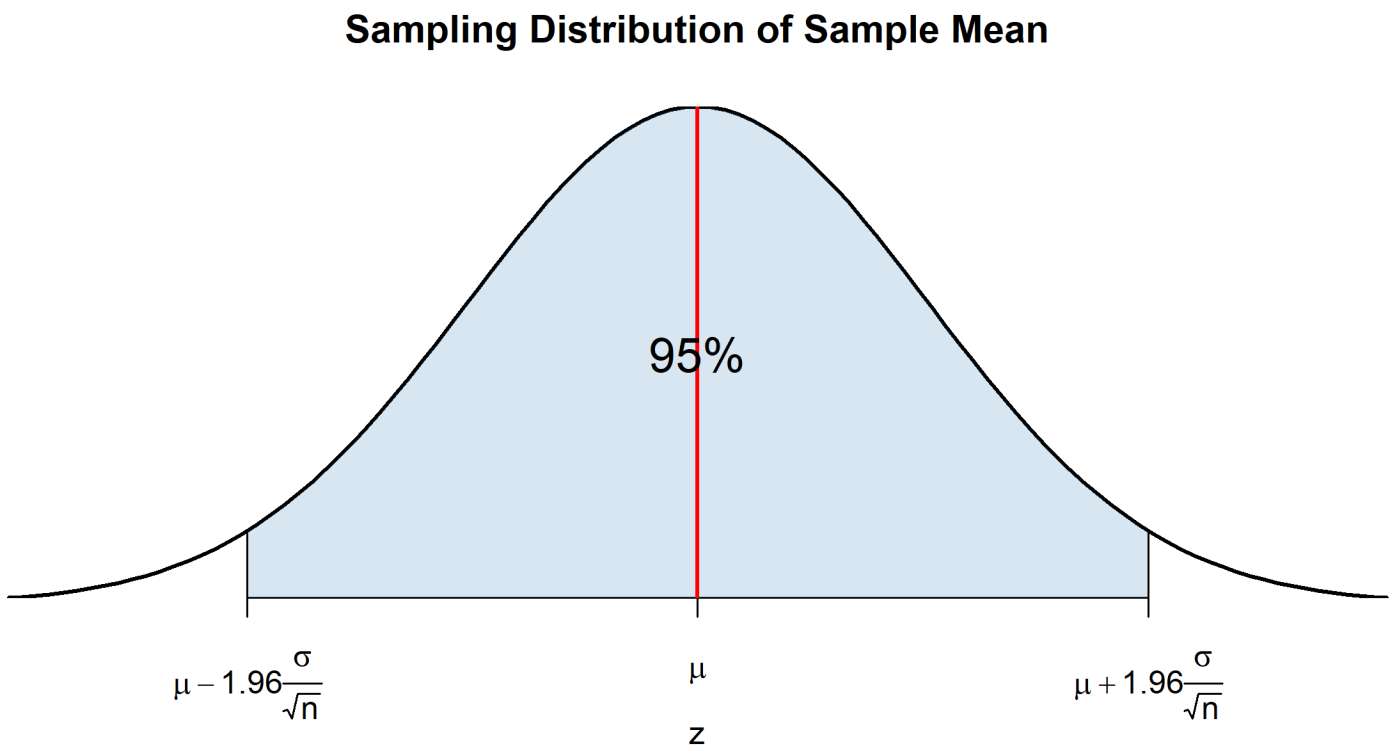
1 · From the sample mean to the margin of error

How far does a sample mean typically sit from the truth? That distance is the building block of every interval.

Sampling distribution

Q: How likely is it that a sample mean is far from the true mean?

- Recall the sampling distribution: $\bar{x} \sim \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$
- Across repeated samples, 95% of the means land in the shaded region. Why 1.96?



Where 1.96 comes from

$$0.95 = P(-1.96 < Z < 1.96) = P\left(z_{\frac{\alpha}{2}} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < z_{1-\frac{\alpha}{2}}\right)$$
$$= P\left(\mu + z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}} < \bar{X} < \mu + z_{1-\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}\right)$$

- The CLT guarantees $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$ is standard normal.
- What if you don't know σ? In large samples $s \to \sigma$, so $\frac{\bar{X}-\mu}{s/\sqrt{n}} \to N(0,1)$.
  - You may need a somewhat larger n so that s approximates σ well.

Margin of error

- So 95% of sample means fall within $1.96\frac{\sigma}{\sqrt{n}}$ of the true mean.
- This half-width is the margin of error: $\mathrm{ME} = z_{1-\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}$.
- Now flip it around. You do not see μ — you see one sample mean $\bar{x}$, and you build the interval $\bar{x} \pm \mathrm{ME}$ around it.

How far can $\bar{x}$ drift from μ before that interval loses the true mean?

Compute the margin of error

A survey of $n = 100$ Airbnb listings has a standard deviation of $s = 961.9$ pesos. Compute the 95% margin of error in pesos: the critical value from `qnorm()`, times the standard error.

Exercise | Start Over | Show Hint | Run Code

```
1 n <- 100
2 s <- 961.9
3 # The 95% margin of error, in pesos
```

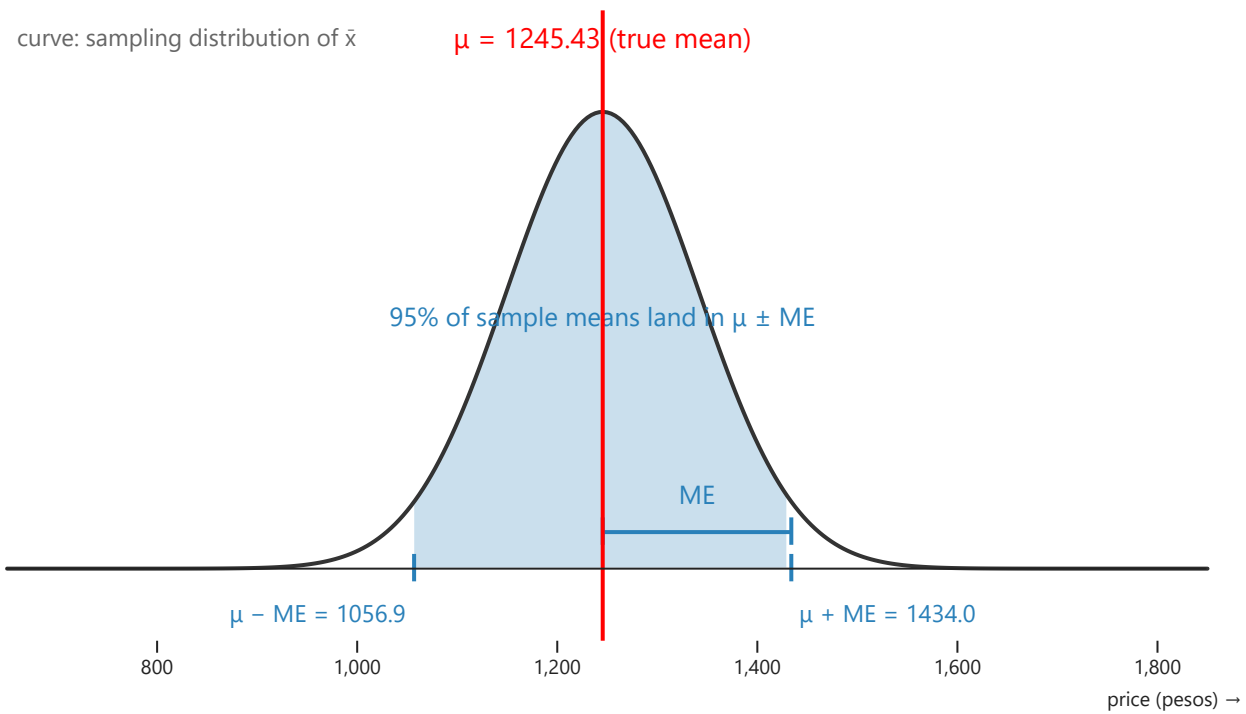
Find the flip point

You just computed it: with $n = 100$ and 95% confidence, $\mathrm{ME} = 188.5$ pesos. Those numbers are fixed for this page — only $\bar{x}$ moves.

Draw a sample mean ☐

Sample mean x̄ (pesos)

1245



SE = σ/√n = **96.2** pesos, ME = 1.96 · SE = **±188.5** pesos.

1. Before touching anything: if $\bar{x}$ lands exactly on the edge of the shaded band — does the interval $\bar{x} \pm \mathrm{ME}$ cover μ, or miss it?
2. Turn on Draw a sample mean. Drag $\bar{x}$ slowly to the right. Write down the price where the segment first turns red.
3. Compare that price with the number at the right edge of the shaded band. Now drag left and check the other side.

What you just found

The flip happens exactly at the edge of the band:

$$|\mu - \bar{x}| < \text{ME} \iff \bar{x} \in (\mu - \text{ME},\ \mu + \text{ME}) \iff \mu \in (\bar{x} - \text{ME},\ \bar{x} + \text{ME})$$

- The interval around $\bar{x}$ covers μ exactly when $\bar{x}$ lands inside the band around μ — the two brackets have the same length, ME.
- And $\bar{x}$ lands inside that band in 95% of samples.
- So the recipe "$\bar{x} \pm \text{ME}$" catches the true mean in 95% of all the samples you could have drawn.

Watch the same story animated: Seeing Theory — confidence intervals.

Does that mean "there is a 95% chance that μ is in *my* interval"? Careful — that question gets its own section (4).

The intuition so far

$$\mathrm{ME} = z_{1-\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}$$

- $s/\sqrt{n}$ is the typical miss: how far a sample mean usually sits from the true mean.
- Multiplying by $z = 1.96$ stretches it to an almost-worst-case miss: only 5% of samples land farther out.
- So an interval of $\pm\mathrm{ME}$ around $\bar{x}$ reaches back to μ — unless your sample was one of the unlucky 5%.
- Precision is bought with data, at a $\sqrt{n}$ rate: 4× the sample halves ME — to halve it again, you need 16×.

2 · Building the interval

The recipe: point estimate, spread, critical value — then $\bar{x} \pm \mathrm{ME}$.

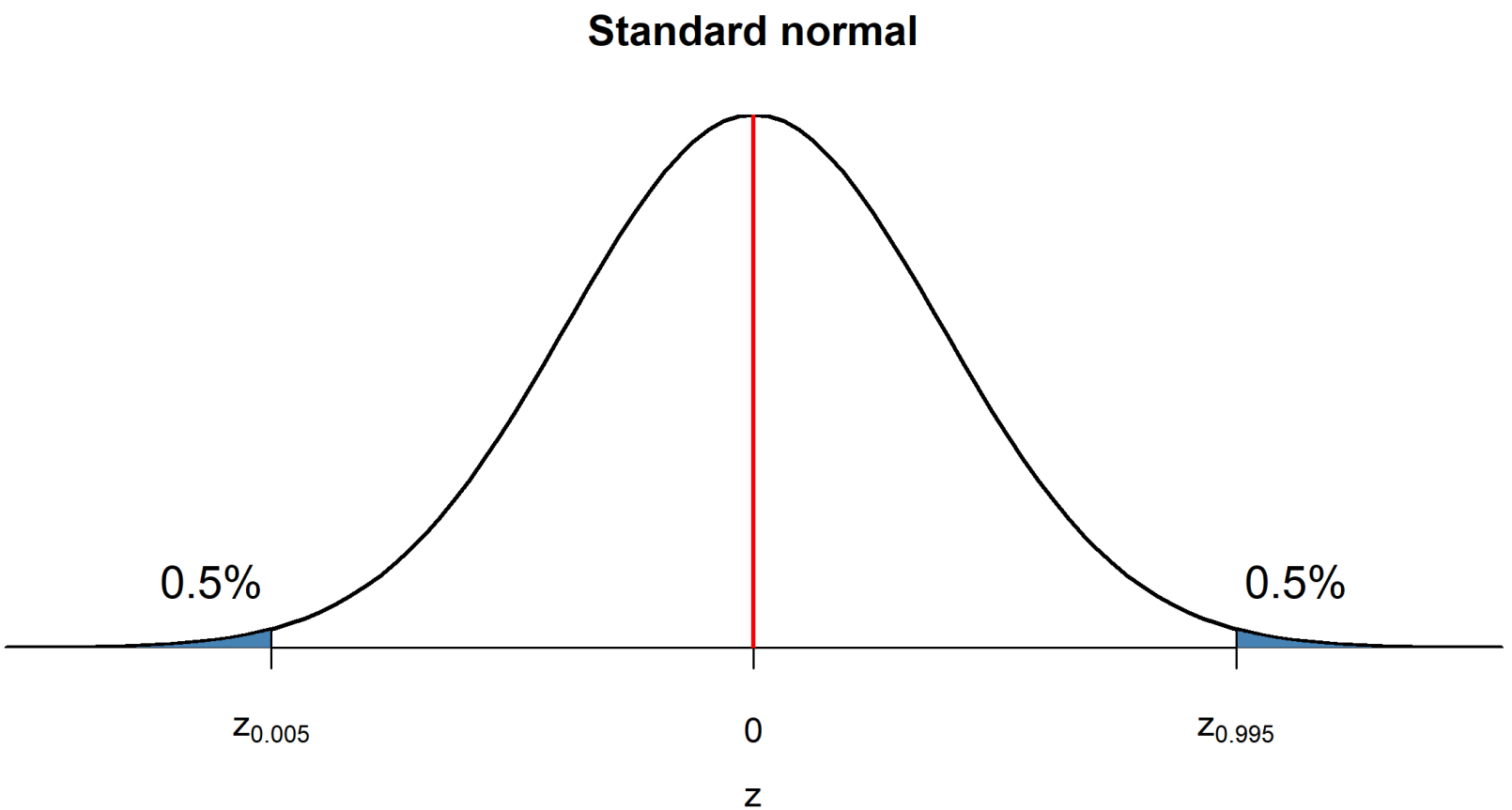
Calculation procedure

Use this when $n > 40$:

1. Take an IID sample.
2. Compute the mean $\bar{x}$ and standard deviation s.
    - Standard error = SD of the estimator: $SE = \frac{s}{\sqrt{n}}$.
3. Pick a confidence level $1 - \alpha$ (usually 90 / 95 / 99%).
    - α = probability of a Type 1 error. Example: 95% → $\alpha = 0.05$.
4. Find critical values $z_{\frac{\alpha}{2}}$, $z_{1-\frac{\alpha}{2}}$ with $P(z_{\frac{\alpha}{2}} < Z < z_{1-\frac{\alpha}{2}}) = 1 - \alpha$.
    - Example: 95% → $z_{0.975} = 1.96$.
5. Build the interval: $\left(\bar{x} - z_{1-\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}, \quad \bar{x} + z_{1-\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}\right)$

Finding critical values

- Suppose the confidence level is 99%, so $\alpha = 0.01$.
- We need $z_{1-\frac{\alpha}{2}}$ with $P(z_{\frac{\alpha}{2}} < Z < z_{1-\frac{\alpha}{2}}) = 0.99$.
- Equivalently $z_{0.995}$ is the 99.5% quantile of the standard normal → $z_{0.995} = 2.58$.



Exercise 1 — Build the 90% interval

You now have all five steps. `Sample_list` holds 200 sampled Airbnb listings: `Sample_list$price` is the nightly price, and `Sample_list$clean` is `TRUE` for the 100 listings with a cleanliness score above 4.5. Build the 90% confidence interval for the mean price of the clean listings, and print both endpoints.

Exercise | Start Over | Show Hint | Run Code

```
1 # The prices of the clean listings first, then steps 2, 4, 5: two endpoints.
```

Constructing a CI: worked example

90% CI for the average price of listings with cleanliness score > 4.5:

1. IID sample, $n = 100$ ✓

2. $\bar{x} = 1245.43$ and $s = 961.9$

3. Confidence level 90% → $\alpha = 0.1$

4. Critical value: $z_{0.95} = 1.65$

5. Interval:

$$\left(1245.43 - 1.65 \cdot \frac{961.9}{\sqrt{100}},\quad 1245.43 + 1.65 \cdot \frac{961.9}{\sqrt{100}}\right) = (1086.64,\ 1404.22)$$

3 · What moves the width

Three dials set how wide the interval comes out. Which ones, and by how much?

Predict before you compute

You are about to rebuild the same interval at 90%, 95% and 99% confidence.

Guess first: how much wider is the 99% interval than the 95% one?

- Roughly the same? 30% wider? Twice as wide?

Commit to a number, then compute it.

Exercise 2 — Change the confidence level

Rebuild the interval for the clean listings' mean price (their prices are in `x`) at 90%, 95% and 99%. Report the three widths, and the ratio of the 99% width to the 95% width.

Exercise | Start Over | Show Hint | Run Code

```r
n  <- length(x)
se <- sd(x) / sqrt(n)

# One interval per confidence level. Which quantile goes with which level?
# ci90 <- ...
# ci95 <- ...
# ci99 <- ...
# then diff() each one, and take the ratio.
```

The price of confidence

Going 95% → 99% costs about 31% extra width — the critical value goes from 1.96 to 2.58 and everything else stays fixed.

- Higher confidence buys a *safer* statement about a *vaguer* range.
- A 100% interval is $(-\infty, \infty)$: always right, never useful.
- Which level you choose is a business decision about the cost of being wrong, not a statistical one.

✏️ Exercise 3 — What does the width actually depend on?

The formula has three inputs you could change: the critical value, n, and s. Hold the confidence level at 95% and find out how the width responds to the other two.

Exercise | Start Over | Show Hint | Run Code

```
ns <- c(10, 25, 50, 100, 400, 1000)

# 1. Width of the 95% interval at each of those n, keeping sd(x) fixed.

# 2. Plot width against n. What shape is it?

# 3. Now keep n = 100 and double the spread: replace sd(x) with 2 * sd(x).
#    What happens to the width?
```

Shape of confidence intervals

$CI = \left(\bar{x} \pm z_{1-\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}\right)$ is wider when:

- the confidence level is higher (99% is wider than 90%);
- the sample size n is smaller;
- the standard deviation σ is larger.

Note the asymmetry you just plotted: s enters linearly (double the spread, double the width) but n enters through $\sqrt{n}$ (quadruple the data, halve the width).

The three dials, live

The same picture as before, now with all three dials unlocked. The sample mean is stuck at $\bar{x} = 1150$.

Confidence level

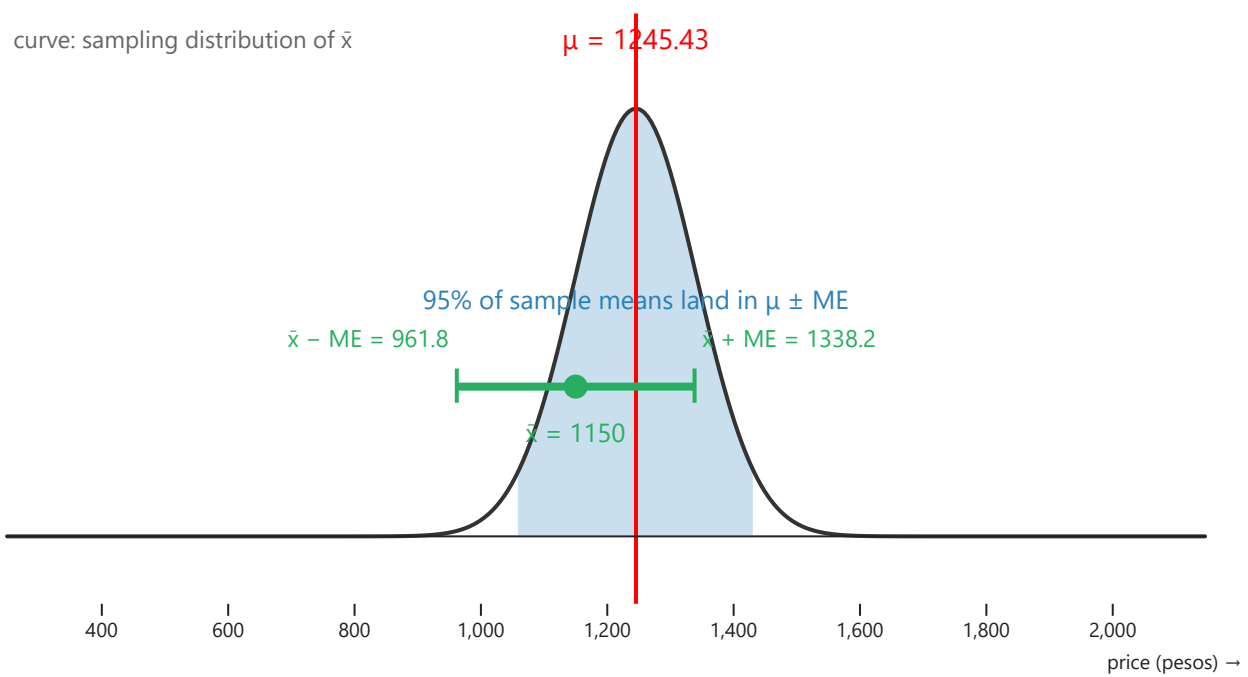
0.95

Sample size n

100

Population sd σ (pesos)

960



z = **1.96**, SE = σ/√n = **96.0** pesos, ME = z · SE = **±188.2** pesos, **width = 2 · ME = 376.3 pesos** — the interval **covers** μ.

1. Double n from 100 to 200, then to 400. How does the width of the interval respond each time?
2. Reset $n = 100$. Push the confidence level from 0.80 up to 0.99. What does the extra confidence cost?
3. Drag σ up: more spread in prices means wider intervals — at every n.
4. At 95%, keep raising n. The interval narrows — and at some point it turns red. Keep that picture in mind for section 4.

4 · What “95% confident” means

You can build an interval. But what exactly is the 95% a probability of?

What does “95% confidence” mean?

You built one interval from one sample. The “95%” is not about that interval — it is about the procedure that produced it.

There is exactly one way to see this: build the interval many times, from a population whose true mean you already know, and count how often it is caught. The Seeing Theory animation does exactly that — watch the running count settle at the confidence level.

Exercise 4 — Say it correctly

Your 95% CI for the mean price is $(980,\ 1130)$. Which statement is the correct interpretation?

a. We are 95% confident the procedure produced an interval that contains the true mean price.

b. There is a 95% probability that the true mean price lies between 980 and 1130.

c. 95% of listing prices fall between 980 and 1130.

d. If we resampled, the new sample mean would be in this interval 95% of the time.

a. is correct.

b. is the one almost everyone picks. In the animation, once an interval is drawn it either contains the true mean or it does not — nothing probabilistic left about it.

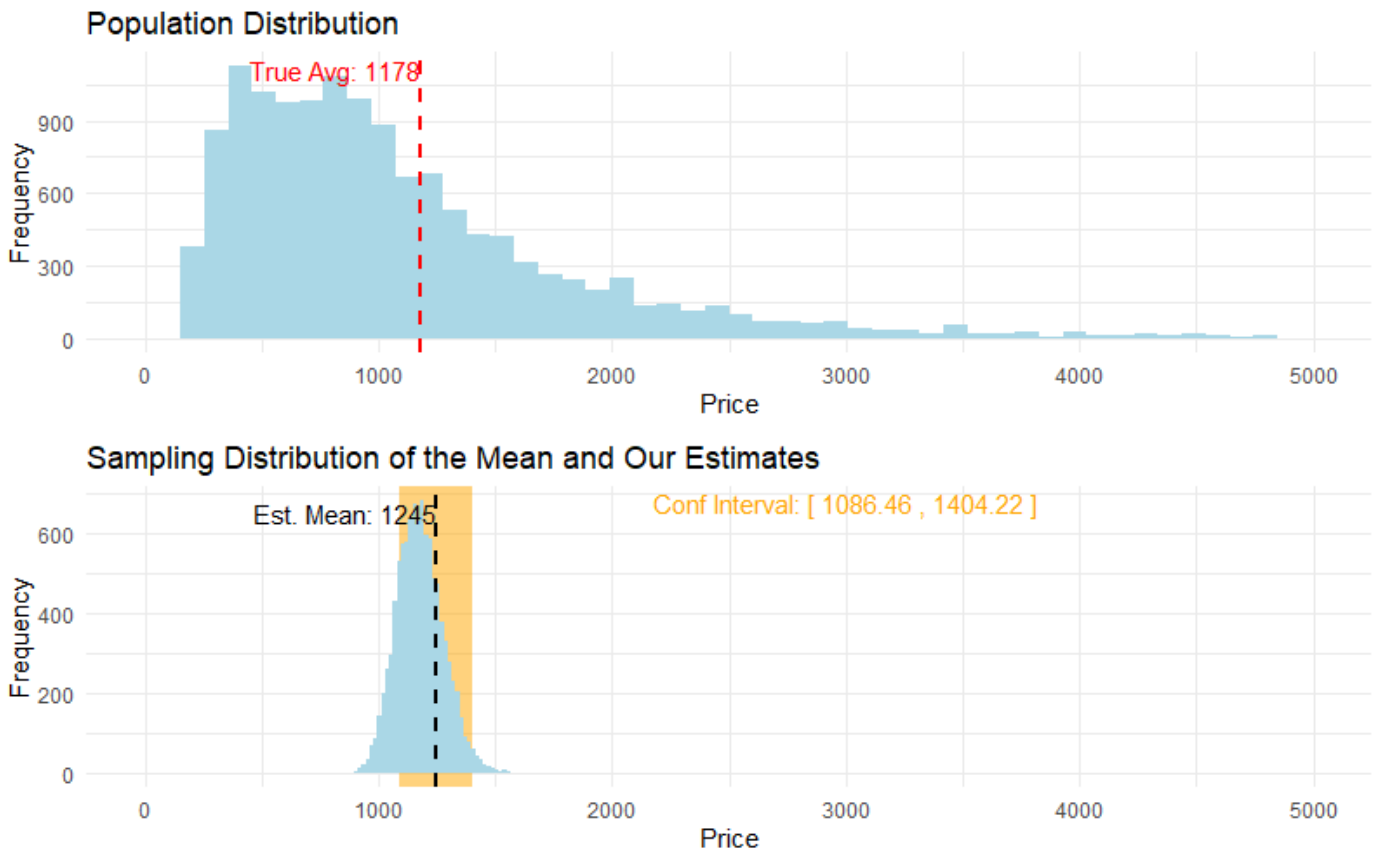
c. confuses the spread of *prices* with the spread of the *mean*: `sd(x)` = 962 vs `se` = 96. (d) describes a prediction interval for a future $\bar{x}$, not a statement about μ.

Interpreting a confidence interval

$$CI_{90} = (1086.64,\ 1404.22)$$

- Correct: we are 90% confident the interval captures the true mean — i.e. that the true mean price is between 1086.64 and 1404.22.
- Incorrect: “with 90% probability the true mean is in this interval.”
  - The computed interval is fixed and the true mean is fixed — no randomness left to attach a probability to.
  - The interval is random only before we draw the sample. After computing, the true mean is either in it or not.

The true Airbnb mean



- Here — exceptionally — we can peek at the answer: the full listing data for the market our 100 clean listings were sampled from.
- Top: every price in that population. The true mean is 1179 pesos.
- Bottom: the sampling distribution of $\bar{x}$, our estimate 1245.43, and our 90% interval.
- The interval caught the true mean: this was one of the lucky 90% of samples. In real work you never get this panel — which is exactly why you report an interval, not a point.

5 · Small samples: Student’s t

With few observations and σ unknown, 1.96 stops being the right multiplier.

When do we use *normal* critical values?

1. X is not normal:

- $n > 30$ — use the normal (use $n > 40$ if σ unknown, so s approximates σ).
- $n < 30$ — normal approximation is unreliable; you need specialized methods (e.g. bootstrap).

2. X is normal:

- Know σ → use the normal, any n (with σ, $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$).
- Don't know σ but $n > 40$ → normal (CLT kicks in).
- Don't know σ and $n < 40$ → use Student's t (s is a poor estimate of σ at low n).

What's Student's t?

If $X_1, \ldots, X_n$ are i.i.d. from $N(\mu, \sigma)$, then

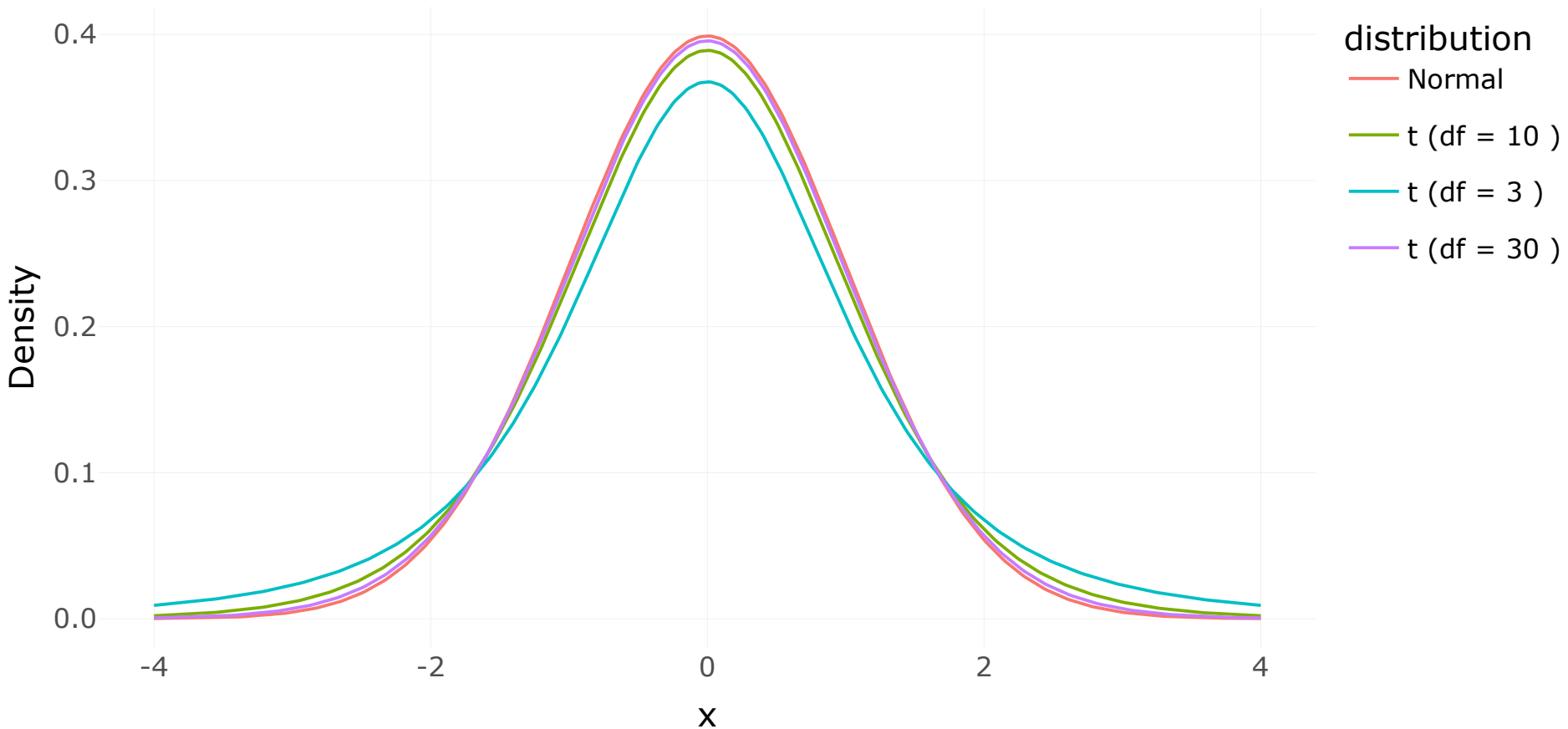
$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

with s the sample standard deviation, follows a Student's t distribution with $n - 1$ degrees of freedom:

$$T \sim t_{n-1}$$

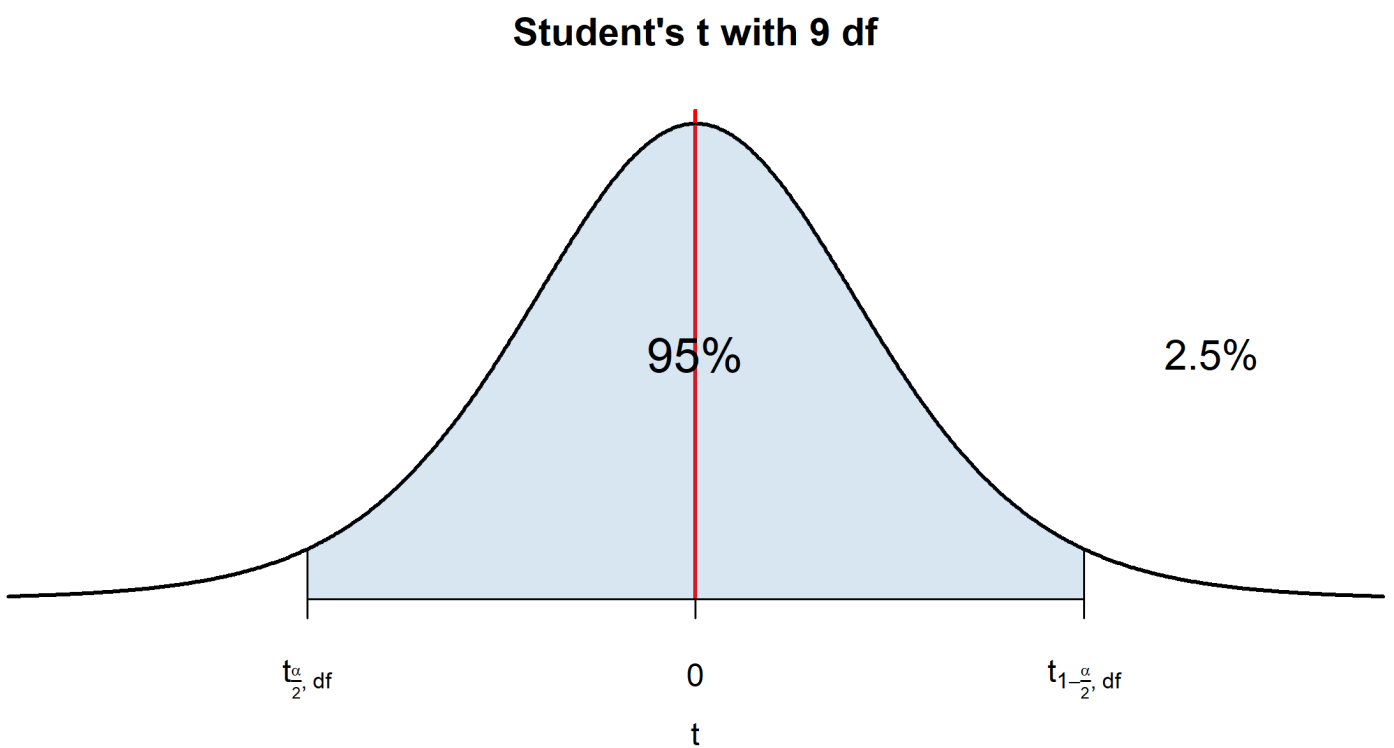
Student's t vs the normal

- Bell-shaped and symmetric around 0.
- Heavier tails than the normal — more uncertainty, because we don't know σ.
- Shape set by the degrees of freedom: as n grows, $t_{n-1} \to N(0,1)$ (CLT again).



Student's t critical values

1. Get the degrees of freedom: $n - 1$.
2. Get the confidence level $1 - \alpha$, hence $\alpha/2$.
3. Find $t_{1-\frac{\alpha}{2},\, n-1}$ with $P(T < t_{1-\frac{\alpha}{2},n-1}) = 1 - \frac{\alpha}{2}$.

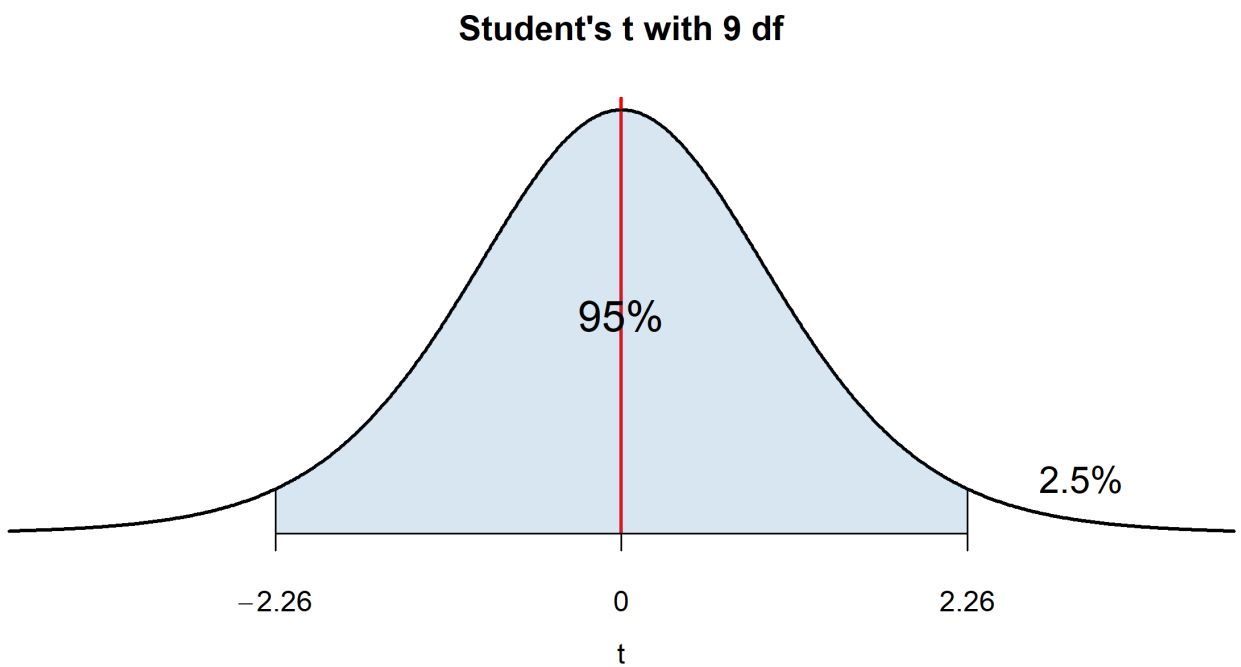


In R: `qt(0.975, 9)` for a 95% CI with 9 df.

t critical value: example

- $n = 10 \rightarrow df = 9$; confidence level 95% → $\frac{\alpha}{2} = 0.025$.
- $t_{0.975,9} = 2.26$, then build the CI as before:

$$\left(\bar{x} - t_{1-\frac{\alpha}{2},n-1} \cdot \frac{s}{\sqrt{n}},\quad \bar{x} + t_{1-\frac{\alpha}{2},n-1} \cdot \frac{s}{\sqrt{n}}\right)$$



The rule says $n < 40$. Where does that come from?

Using `qnorm` when you should have used `qt` makes your interval too narrow.

At what sample size does the mistake stop mattering? Pick a number before you find out.

Exercise 5 — Find the crossover

Compute both 95% critical values — `qt(0.975, n - 1)` and `qnorm(0.975)` — for a range of sample sizes, and report how far apart they are in percent. Find the smallest `n` in your grid where the gap drops below 5%, and below 1%.

Exercise | Start Over | Show Hint | Run Code

```
ns <- c(2, 5, 10, 20, 30, 40, 50, 100, 500, 1000)

# t critical value at each n, the (single) normal critical value,
# and the percentage by which t exceeds z.
```

Where the crossover lands

- Below $n \approx 20$ the difference is large and t is not optional.
- By $n = 30$ the t value is ~4% above z; by $n = 40$, ~3%; by $n = 100$, ~1%.
- That is the whole justification for the $n > 40$ rule of thumb — nothing deeper.

But: using `qt` is never wrong. It converges to `qnorm` by itself. When σ is unknown, defaulting to t costs you nothing.

Exercise 6 — A small sample

Take only the first 12 prices, `xs <- x[1:12]`, and build a 95% interval for the mean. Build it twice — once with `qnorm`, once with `qt` — and report both widths.

Exercise | Start Over | Show Hint | Run Code

```
1  xs <- x[1:12]
2
3  # Same recipe, on xs. Which critical value belongs here, and what does the other one cost you?
```

Exercise 7 — Spending with free shipping

Free shipping was offered to a random group of customers. Spend per customer ($n = 10$):

$$157.80,\ 192.45,\ 210.20,\ 175.60,\ 198.30,\ 180.90,\ 205.75,\ 185.20,\ 177.40,\ 195.60$$

a. Compute the 90% CI. Which critical value, and what must you assume?

b. Average spending without free shipping is $182. Is 182 inside your interval? What does that say about the plausible values of the mean with free shipping?

Exercise | Start Over | Show Hint | Run Code

```r
spend <- c(157.80, 192.45, 210.20, 175.60, 198.30,
           180.90, 205.75, 185.20, 177.40, 195.60)

# (a) 90% interval

# (b) is 182 inside it?
```

Exercise 8 — The whole week, one question

You are the analyst for a host with listings in this market. The city tourism board has published a report stating that

> *"the average nightly price in this market is 900 pesos."*

Your sample is `x` — the nightly prices of the 100 clean listings. Answer with the tools of this week: a number, a picture, and one sentence.

Exercise | Start Over | Show Hint | Run Code

```
# 1. A number: the range of values for the true mean that your sample supports.
#    Which critical value, and why that one at this n?

# 2. A picture: where the data sit -- then read your interval and 900
#    against its axis.

# 3. One sentence: what do you tell the host about the board's claim?
```

Exam question — Fall 2025 Midterm 1

Real questions from last year's Midterm 1 — Households & Environment Survey ($n = 14{,}505$ households, each surveyed once, 2017). The data frame `exam_data` is loaded below — solve them right here.

Q7. A transportation agency wants to estimate average weekly household gasoline spending, `exam_data$gasoline_spending_weekly`. Construct a 90% confidence interval for the mean and report the lower bound.

Q8. What happens to the interval if you change the confidence level to 99%? Answer first, then check with code.

Exercise | Start Over | Show Hint | Run Code

```
1 # Q7: the 90% interval, then its lower bound.
2
3 # Q8: say what changes first -- then rebuild the interval at 99% and check.
```

Practice now (this week's pack): the CI-construction and confidence-level problems in the Week 5 set, then re-run Q7 on `exam_data`.

→ More exam practice at the end of this deck

What you must be able to do this week

- Construct a confidence interval for a mean — compute $\bar{x}$, s, the standard error, and the margin of error, then $\bar{x} \pm \mathrm{ME}$.
- Choose the right critical value: normal (`qnorm`, large n) vs Student's t (`qt`, small n, σ unknown).
- Interpret a CI correctly — confidence about the procedure, *not* a probability about a fixed interval.
- Explain what makes an interval wider or narrower: confidence level, sample size, and spread.
- Build a confidence interval for a variance with the (asymmetric) chi-square quantiles (`qchisq`).

Next: from estimating parameters to testing claims about them — hypothesis testing.

Practice for this week

Weekly exercise set: `W05_exercises.pdf`, with solutions in `W05_solutions.pdf`.

You can already solve exercises 1–12.

Suggested before the next class, closest to the exam:

Exercise	Topic	How
3	Productivity test of 25 employees: interval with the t distribution	R as a calculator
6	Defective computers: interval for a proportion	R as a calculator
8	Streaming hours: minimum sample size	R as a calculator
9	Union membership by gender (`cps`)	R with the data
10	Strike durations (`strikes`)	R with the data

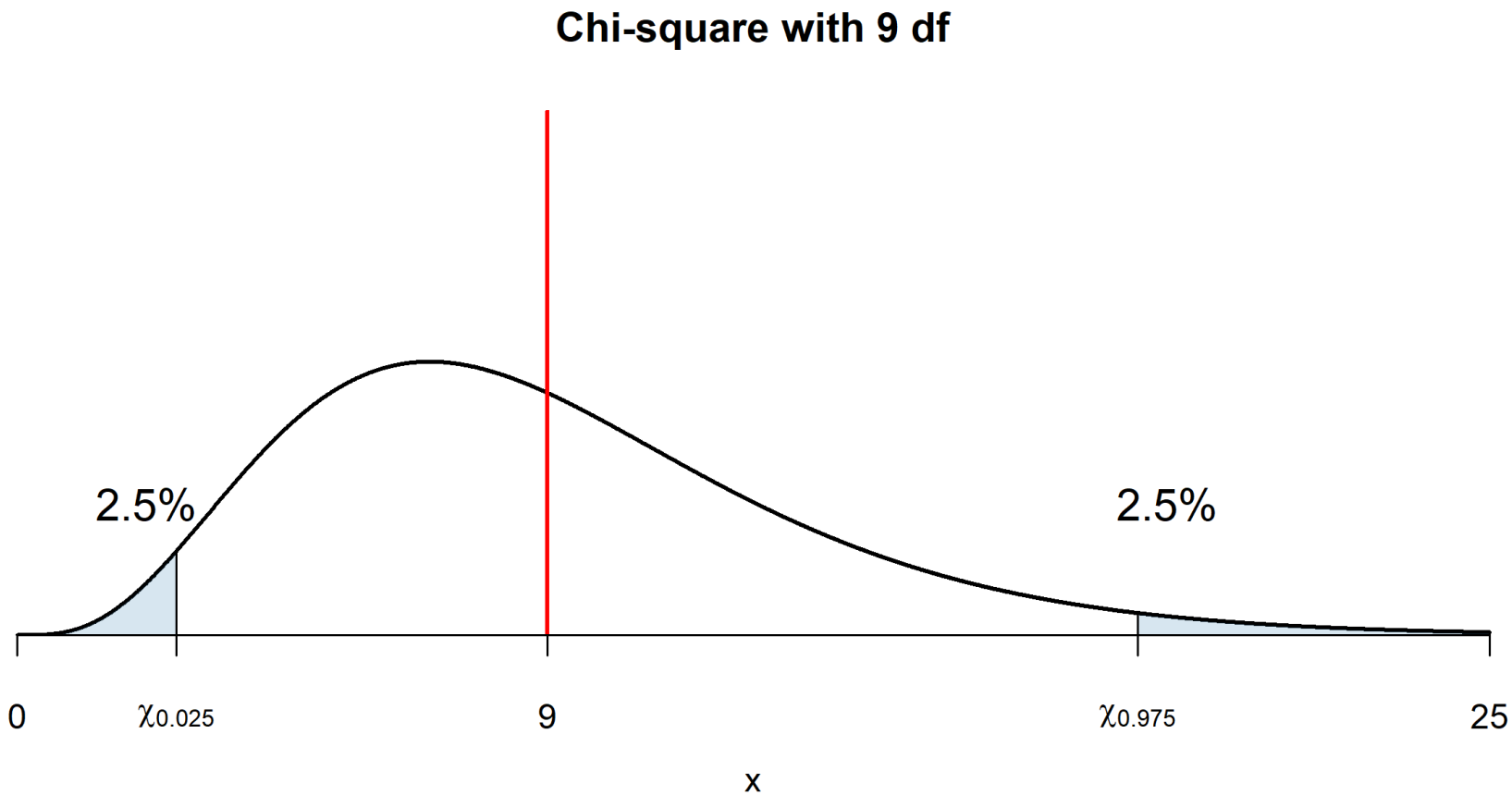
Appendix — optional theory refresher

Confidence intervals for variance

- Sometimes we care about spread, not the mean.
- A machine making bolts with high diameter variance is unreliable even if the average is correct.

- For $X_1, \ldots, X_n$ from a normal, the sample variance $S^2 = \frac{\sum_i (X_i - \bar{X})^2}{n-1}$ satisfies:

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}, \qquad P\left(\chi^2_{0.025,n-1} < \frac{(n-1)S^2}{\sigma^2} < \chi^2_{0.975,n-1}\right) = 0.95$$



Where does this distribution come from? The chi-square, built from the normal.

CI for variance: construction

Invert the chi-square statement to isolate σ^2:

$$\begin{aligned} 0.95 &= P\left(\chi^2_{0.025,n-1} < \frac{(n-1)S^2}{\sigma^2} < \chi^2_{0.975,n-1}\right) \\ &= P\left(\frac{(n-1)S^2}{\chi^2_{0.975,n-1}} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{0.025,n-1}}\right) \end{aligned}$$

So, generally:

$$CI_{1-\alpha} = \left(\frac{(n-1)S^2}{\chi^2_{1-\frac{\alpha}{2},n-1}},\ \frac{(n-1)S^2}{\chi^2_{\frac{\alpha}{2},n-1}}\right)$$

- The χ^2 quantiles come from R via `qchisq(alpha, df)`. Note the interval is asymmetric (chi-square is skewed).

Bonus exercise — Variance CI with `qchisq()`

Quality control: you measure fat content in sausages. From a random sample of 12 sausages you find a sample variance of 20 grams^2. Build the 99% confidence interval for the population variance, and state what you had to assume.

Exercise | Start Over | Show Hint | Run Code

```r
n  <- 12
s2 <- 20
# Step 1: the two critical values with qchisq()

# Step 2: the interval -- which quantile belongs in the LOWER bound?

# Step 3: is the interval symmetric around s2 = 20? Check the two half-widths.
```

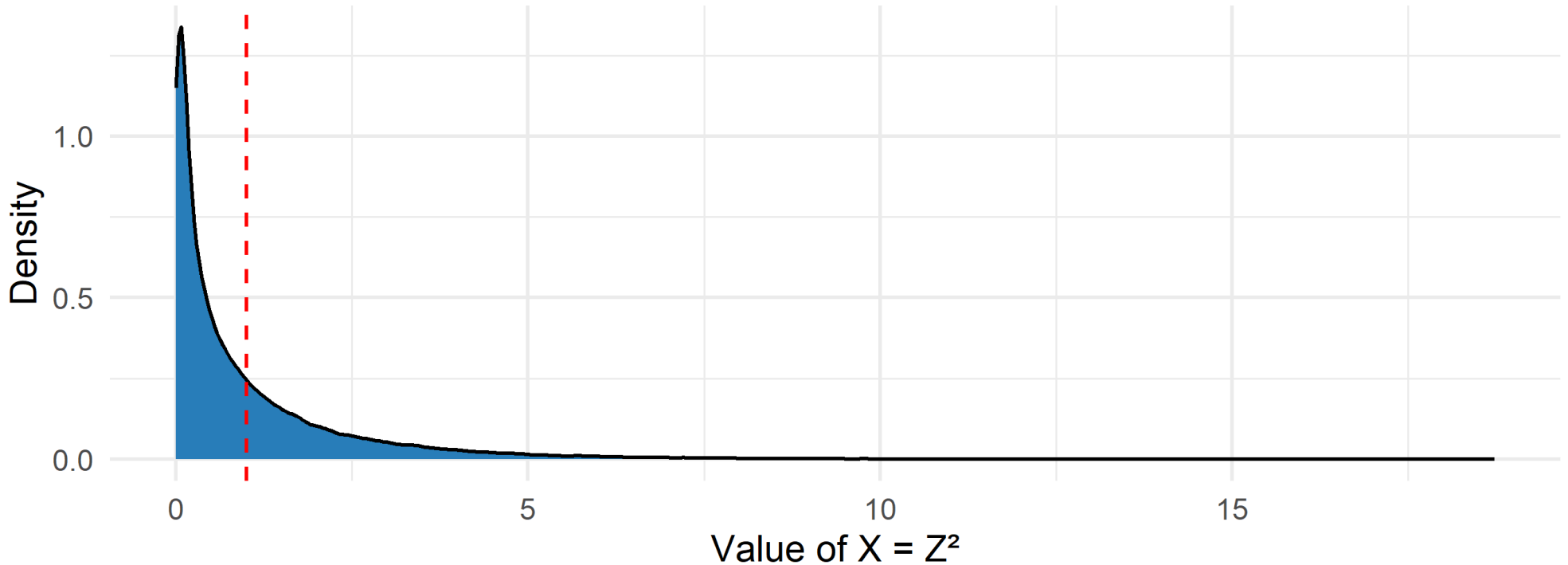
Where the chi-square comes from

- Sample means are normal in large samples. What about the sample standard deviation?
- If we take many samples and compute each one's SD, what does *that* distribution look like?

- If X_i is normal, the sample variance is related to the chi-square distribution.
- That is the distribution the variance interval is built on.

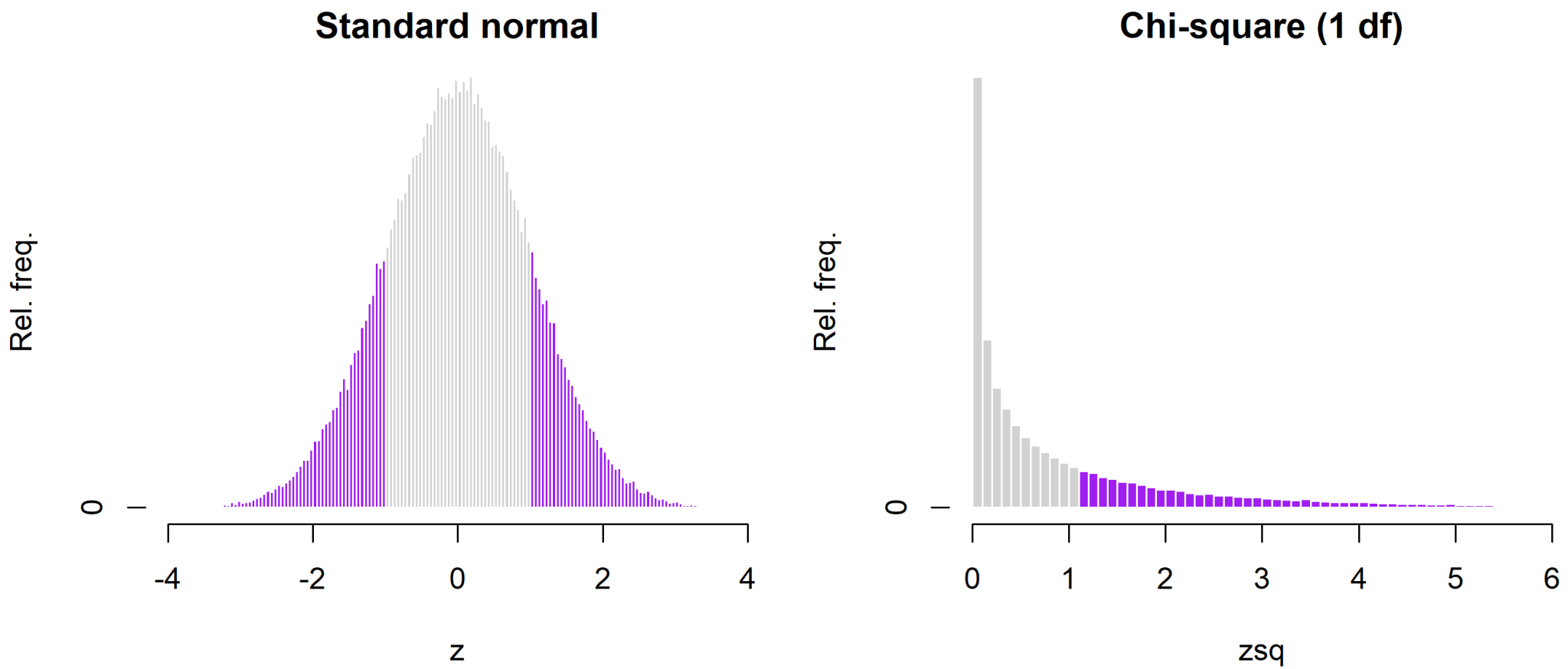
From normal to chi-square

- Start with the standard normal $Z \sim N(0,1)$. The transformation $X = Z^2$ gives χ^2_1.
- $E[\chi^2_1] = Var(Z) = 1; Var(\chi^2_1) = 2.$



Seeing the connection

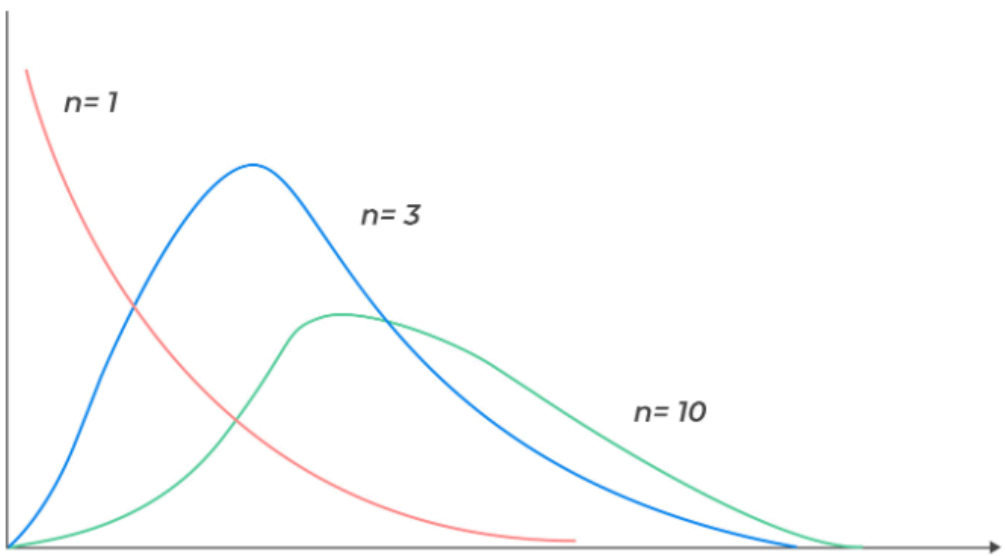
Shaded area = $P(X = Z^2 > 1)$ where $X \sim \chi^2_1, Z \sim N(0,1)$ — the same area in both panels:



Chi-square and the sum of normals

- The sum of n i.i.d. squared standard normals is χ^2_n:

$$\sum_{i=1}^{n} Z_i^2 \sim \chi^2_n, \qquad E[\chi^2_n] = n, \qquad Var(\chi^2_n) = 2n$$



- Why does the shape approach the normal for large n? It is a sum of random variables — the CLT again.

← Back to the variance interval

Exam practice

These problems mirror this week's exercise set and the exam — solve each one right here (the Solution button reveals the full worked solution).

← Back to the exam spotlight

Exam practice 1 — Stock returns CI

You are considering investing in a Mexican company listed on the *Bolsa Mexicana de Valores*. Its annual returns over the last 10 years are: $0.03,\ -0.12,\ 0.15,\ 0.04,\ -0.14,\ -0.06,\ 0.00,\ 0.08,\ 0.20,\ 0.07$. Assume returns are i.i.d. normal. [I3]

What is the 95% CI for μ?

Exercise | Start Over | Show Hint | Run Code

```r
returns <- c(0.03, -0.12, 0.15, 0.04, -0.14, -0.06, 0.00, 0.08, 0.20, 0.07)

# The 95% interval
```

Exam practice 2 — Government bond returns

Monthly returns on a Mexican government bond are i.i.d. from a normal distribution with unknown mean and standard deviation. You observe 12 months, with sample average 0.012 and standard deviation 0.07. [I11]

a. Provide 95% and 99% confidence intervals for μ_X using the t distribution.

b. Compare these intervals to those based on $N(0, 1)$.

c. Suppose instead $n = 100$, same sample average and standard deviation. Provide 95% and 99% CIs using the normal approximation.

Exercise | Start Over | Show Hint | Run Code

```
1 xbar <- 0.012; s <- 0.07; n <- 12
2
3 # (a) two t-based intervals
4
5 # (b) the same two with normal critical values -- which are wider, and by how much?
6
7 # (c) n = 100, normal approximation
```

Exam practice 3 — Employee productivity

A random sample of 25 employees at a firm in Mexico City yields $\bar{x} = 72$ (standardized productivity test) and $s_x = 15$. Assume test scores are normally distributed with unknown mean and variance. [I12]

a. Provide 90% and 95% confidence intervals for μ_X.

b. Compare these to intervals based on $N(0, 1)$.

Exercise | Start Over | Show Hint | Run Code

```
1  xbar <- 72; s <- 15; n <- 25
2
3  # (a) 90% and 95%, t-based
4
5  # (b) the same two, normal-based
```

c. If instead $\bar{x} = 76$, $s_x = 15$, how would the width of the 95% CI compare?

Unchanged — width $= 2 \times 2.064 \times 3 = 12.38$. Changing $\bar{x}$ shifts the interval's *center* but not its width; width depends only on s, n, and the critical value.

d. If instead $\bar{x} = 72$, $s_x = 20$, how would the width of the 95% CI compare?

Wider — $SE = 20/5 = 4$, width $= 2 \times 2.064 \times 4 = 16.51$. Greater variability means more uncertainty about the population mean.

Exam practice 4 — Reverse-engineer the CI

A Mexican company's weekly profits are i.i.d. draws from $X \sim N(\mu, \sigma^2)$. After n weeks, two confidence intervals for μ are constructed — 95% CI: $(9.201,\ 14.799)$ and 90% CI: $(9.955,\ 14.045)$. [I13]

a. What is the sample average of weekly profits?

b. What is n? (assume the Student's t distribution was used)

c. What is the sample standard deviation of weekly profits?

Exercise | Start Over | Show Hint | Run Code

```
1  # (a) what feature of an interval is the sample average?
2
3  # (b) the two intervals share a center and a SE -- only the critical value differs.
4  #     Search over df: which one reproduces the ratio of half-widths?
5
6  # (c) back out s from the half-width
```

📝 Exam practice 5 — Rescale a confidence interval

The two-sided asymptotic 95% confidence interval for a parameter is $(12,\ 22)$. [I15]

a. What is the two-sided asymptotic 80% confidence interval for the same parameter?

Exercise | Start Over | Show Hint | Run Code

```
1 # The interval is all you get. Recover its two ingredients first, then rebuild.
```

Exam practice 6 — Defective computers (proportion)

A university IT department in Mexico has many old computers, some no longer working. To estimate the proportion defective, staff test 80 randomly and find 20 defective. Find the asymptotic 95% CI for the population proportion. [I16]

Exercise | Start Over | Show Hint | Run Code

```
1 n <- 80; x <- 20
2
3 # A proportion is a mean of a 0/1 variable -- but its SE has its own formula.
```

Exam practice 7 — Gardening promotion

A gardening company offers a promotion: n households get a free first-time service, after which some continue as paying customers. The probability of continuing is π. After observation, the asymptotic 95% confidence interval for π is $(0.1800,\ 0.3200)$. [I17]

a. What is the sample proportion?

b. What is n?

Exercise | Start Over | Show Hint | Run Code

```
1  # (a) read the sample proportion off the interval
2
3  # (b) the half-width contains n -- invert the proportion-SE formula to get it out
```

Exam practice 8 — Sample size for a target width

A marketing firm wants to estimate the average daily hours of streaming (Netflix/Disney+) per household in Mexico. Suppose $\sigma = 2.5$ hours. They want the 99% confidence interval width to be no larger than 0.6 hours. What is the minimum required sample size? [I18]

Exercise | Start Over | Show Hint | Run Code

```
1 sigma <- 2.5
2
3 # Write the width as a function of n, set it equal to the target, solve for n.
4 # Watch two things: width vs half-width, and rounding in the right direction.
```

📝 Exam practice 9 — Union membership by gender

The `cps` dataset (package `probstats4econ`) contains U.S. labor-force data with `gender` and `unionstatus`. The two-way table `table(cps$gender, cps$unionstatus)` is loaded here as `cps_tab`. [I19]

a. Get the union rate within each gender out of the table.

b. Compute the asymptotic 95% confidence interval for the probability that a male worker is in a union.

c. Compute the asymptotic 95% confidence interval for the probability that a female worker is in a union.

Exercise | Start Over | Show Hint | ▷ Run Code

```
1 cps_tab
2
3 # (a) the raw counts are not comparable -- there are more men. Turn each ROW into shares.
4
5 # (b), (c) each rate is a proportion with its own n: the ROW total.
```

Exam practice 10 — Strike durations

The `strikes` dataset (package `probstats4econ`; 566 strike durations in weeks, 1968–1976) has key variable `duration`. Part (a) asks for the mean, median, and standard deviation — at home run `mean(strikes$duration)` etc.; the results are: mean = 43.624, median = 28, sd = 44.666. [I20]

b. Construct a 90% confidence interval for the average strike duration.

Exercise | Start Over | Show Hint | Run Code

```
1 xbar <- 43.624; s <- 44.666; n <- 566
2
3 # The 90% interval. Which critical value at this n?
4
5 # Then: the interval is a few weeks wide, but sd is 44.7 weeks.
6 # Compare the two numbers -- what is each one describing?
```

The mean (43.6) is well above the median (28). What does that tell you about the distribution?

Right-skewed — a few very long strikes pull the mean up above the median.